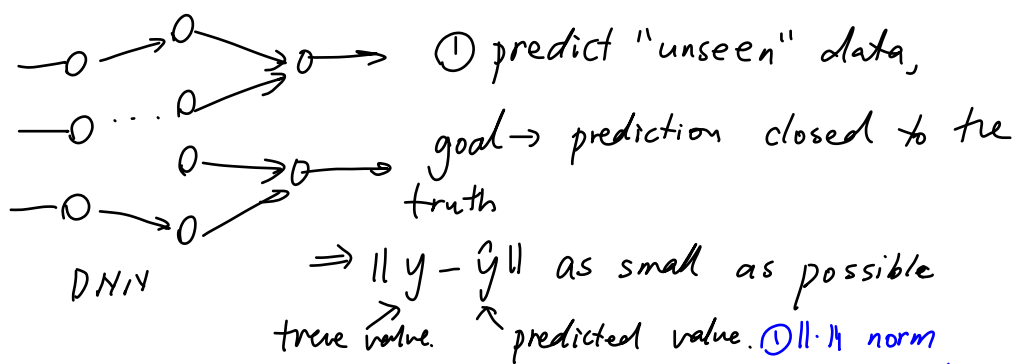


Analytic Geometry I: Norm, Inner Product and Positive definite

Data set
input, output



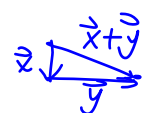
① $\|\cdot\|$ norm
② distance between y and $\hat{y}$

1. Norm

Def. A norm on a vector space V is a function

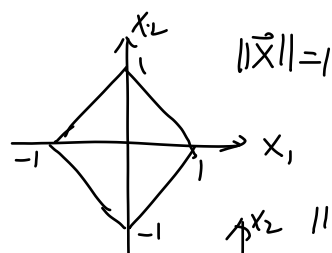
$$\|\cdot\|: V \mapsto \mathbb{R}$$

such that for any $\lambda \in \mathbb{R}$ and $\vec{x}, \vec{y} \in V$, s.t. the following hold:

- (1) $\|\lambda \vec{x}\| = |\lambda| \cdot \|\vec{x}\|$ absolute homogeneous
- (2) $\|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\|$ triangle inequality 
- (3) $\|\vec{x}\| \geq 0$, and $\|\vec{x}\| = 0 \iff \vec{x} = \vec{0}$: positive definite

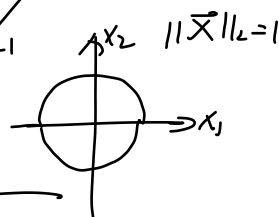
Example: (1) ℓ_1 -norm (Manhattan Norm)

$$\text{For } x \in \mathbb{R}^n \quad \|\vec{x}\|_1 := \sum_{i=1}^n |x_i|$$



(2) ℓ_2 -norm (Euclidean Norm)

$$\text{For } x \in \mathbb{R}^n, \quad \|\vec{x}\|_2 = \sqrt{\sum_{i=1}^n x_i^2} = \sqrt{\vec{x}^T \vec{x}}$$



2. General Inner product

① Dot product: for $\vec{x}, \vec{y} \in \mathbb{R}^n$, $\vec{x}^T \vec{y} = \sum_{i=1}^n x_i y_i$

② Bilinear mapping f

Def. Given a vector space V . For all $\vec{x}, \vec{y} \in V$, $\lambda, \varphi \in \mathbb{R}$,

$f(\cdot, \cdot): V \times V \mapsto \mathbb{R}$ is a bilinear map

1) $f(\lambda \vec{x} + \varphi \vec{y}, \vec{z}) = \lambda f(\vec{x}, \vec{z}) + \varphi f(\vec{y}, \vec{z})$ (linear in 1st argument)

2) $f(\vec{x}, \lambda \vec{y} + \varphi \vec{z}) = \lambda f(\vec{x}, \vec{y}) + \varphi f(\vec{x}, \vec{z})$ (linear in 2nd argument)

③ Symmetric.

Let V be a vector space and $f: V \times V \mapsto \mathbb{R}$ be a bilinear

mapping. Then f is symmetric if $f(\vec{x}, \vec{y}) = f(\vec{y}, \vec{x})$
 $\Rightarrow$ the order of argument does not matters, (x & y)

④ positive definite.

Let V be a vector space and $f: V \times V \rightarrow \mathbb{R}$ be a ~~bilinear~~ mapping

Then f is positive definite if $\forall \vec{x} \in V \setminus \{\vec{0}\}$, we have

$$f(\vec{x}, \vec{x}) > 0 \text{ and } f(\vec{0}, \vec{0}) = 0 \text{ (always non-negative)}$$

(General) Inner product

Def. Consider a vector space V and a mapping $f: V \times V \rightarrow \mathbb{R}$

f is called an inner product on V , if.

① f is bilinear ; ② f is symmetric ; ③ f is positive definite

In this case, we write $f(\vec{x}, \vec{y})$ as $\langle \vec{x}, \vec{y} \rangle$

3. Property of Inner Product

Consider a vector space V , inner product $\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{R}$

and an ordered basis $B = [\vec{b}_1, \dots, \vec{b}_n]$

For any $\vec{x}, \vec{y} \in V$, with

$$\vec{x} = \sum_{i=1}^n \psi_i \vec{b}_i, \quad \vec{y} = \sum_{j=1}^n \lambda_j \vec{b}_j, \text{ with suitable } \psi_i, \lambda_j \in \mathbb{R}$$

$$\text{Then } \langle \vec{x}, \vec{y} \rangle = \left\langle \sum_{i=1}^n \psi_i \vec{b}_i, \sum_{j=1}^n \lambda_j \vec{b}_j \right\rangle$$

$$= \sum_{i=1}^n \psi_i \left\langle \vec{b}_i, \sum_{j=1}^n \lambda_j \vec{b}_j \right\rangle = \sum_{i=1}^n \psi_i \sum_{j=1}^n \lambda_j \langle \vec{b}_i, \vec{b}_j \rangle$$

$$= \sum_{i=1}^n \sum_{j=1}^n \psi_i \langle \vec{b}_i, \vec{b}_j \rangle \lambda_j = \hat{x}^T A \hat{y}$$

$$\text{with } \hat{x} = [\psi_1, \dots, \psi_n]^T, \quad \hat{y} = [\lambda_1, \dots, \lambda_n]^T = A = \begin{bmatrix} \langle \vec{b}_1, \vec{b}_1 \rangle & \dots & \langle \vec{b}_1, \vec{b}_n \rangle \\ \langle \vec{b}_2, \vec{b}_1 \rangle & \dots & \dots \\ \vdots & \ddots & \vdots \\ \langle \vec{b}_n, \vec{b}_1 \rangle & \dots & \langle \vec{b}_n, \vec{b}_n \rangle \end{bmatrix}$$

A is symmetric since $\langle \vec{b}_j, \vec{b}_i \rangle = \langle \vec{b}_i, \vec{b}_j \rangle$.

Example: Consider $V = \mathbb{R}^2$ with an inner product

$\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{R}$, and an order basis $B = [\vec{q}_1, \vec{q}_2]$ of V

where $\vec{q}_1 = [1, 1]^T$, $\vec{q}_2 = [1, -2]^T$. Given $\vec{x} = 2\vec{q}_1 + 3\vec{q}_2$

$\vec{y} = -\vec{q}_1 + 2\vec{q}_2$, Compute $\langle \vec{x}, \vec{y} \rangle$, (dot product)

Normal Computations $\langle \vec{x}, \vec{y} \rangle = \langle 2\vec{q}_1 + 3\vec{q}_2, -\vec{q}_1 + 2\vec{q}_2 \rangle$

$$= -2\langle \vec{q}_1, \vec{q}_1 \rangle - 3\langle \vec{q}_2, \vec{q}_1 \rangle + 4\langle \vec{q}_1, \vec{q}_2 \rangle + 6\langle \vec{q}_2, \vec{q}_2 \rangle$$

$$\langle \vec{q}_1, \vec{q}_1 \rangle = 2, \quad \langle \vec{q}_2, \vec{q}_1 \rangle = \langle \vec{q}_1, \vec{q}_2 \rangle = -1, \quad \langle \vec{q}_2, \vec{q}_2 \rangle = 5$$

$$\Rightarrow \langle \vec{x}, \vec{y} \rangle = -2 \times 2 - 3 \times (-1) + 4 \times (-1) + 6 \times 5 = 25.$$

$$\text{also } A = \begin{bmatrix} \langle \vec{q}_1, \vec{q}_1 \rangle & \langle \vec{q}_1, \vec{q}_2 \rangle \\ \langle \vec{q}_2, \vec{q}_1 \rangle & \langle \vec{q}_2, \vec{q}_2 \rangle \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 5 \end{bmatrix}$$

$$\vec{x} = [2, 3]^T, \quad \vec{y} = [-1, 2]^T$$

$$\vec{x}^T A \vec{y} = [2 \quad 3] \begin{bmatrix} 2 & -1 \\ -1 & 5 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = 25$$

7. Symmetric and Positive Definite Matrix

Recall the positive definite of a norm, we should have.

$\forall \vec{x} \in V \setminus \{\vec{0}\}$, $\vec{x}^T A \vec{x} > 0$, we call a symmetric matrix A satisfying this condition as positive definite matrix.

Moreover, if $\forall \vec{x} \in V \setminus \{\vec{0}\}$, $\vec{x}^T A \vec{x} \geq 0$, then A is called symmetric, positive semidefinite.

Theorem 1: For any $A \in \mathbb{R}^{n \times n}$, $A^T A$ is positive semidefinite

Proof: by definition, for any $\vec{x} \in V$, $A\vec{x} \in V$ $\vec{x}^T A^T A \vec{x} \geq 0$
 $\Rightarrow A^T A \geq 0$

Theorem 2. For any $M \in \mathbb{R}^{n \times n}$ that is symmetric, positive definite, iff there exists $A \in \mathbb{R}^{n \times n}$ with full rank, s.t.

$$M = A^T A$$

Theorem 3. For a real-valued, finite-dimensional vector space V and an ordered basis B of V , it holds that:

$\langle \cdot, \cdot \rangle : V \times V \mapsto \mathbb{R}$ is an inner product if and only if there exists a symmetric, positive definite matrix $A \in \mathbb{R}^{n \times n}$

with $\vec{x}, \vec{y} \in V$ $\langle \vec{x}, \vec{y} \rangle = \hat{x}^T A \hat{y}$, where $\hat{x}$ and $\hat{y}$ are the coordinate representations of $\vec{x}, \vec{y}$ with respect to B .

Exercise: Consider a positive definite matrix $M \in \mathbb{R}^{n \times n}$, with a vector space $V = \mathbb{R}^n$, prove

$\forall \vec{x} \in V$, $\|\vec{x}\|_M := \sqrt{\vec{x}^T M \vec{x}}$ is a norm.

Proof: (Condition 1). Consider $\lambda \in \mathbb{R}$, $\vec{x} \in V$

$$\|\lambda \vec{x}\|_M = \sqrt{\lambda^2 \vec{x}^T M \vec{x}} = |\lambda| \sqrt{\vec{x}^T M \vec{x}} = |\lambda| \cdot \|\vec{x}\|_M \quad \checkmark$$

(Condition 2) Consider any $\vec{x}, \vec{y} \in V$

$$\|\vec{x} + \vec{y}\|_M \leq \|\vec{x}\|_M + \|\vec{y}\|_M$$

$$\|\vec{x} + \vec{y}\|_M = \sqrt{(\vec{x} + \vec{y})^T M (\vec{x} + \vec{y})}$$

$$\vec{x}^T M^T M \vec{y}$$

$$= \sqrt{\vec{x}^T M \vec{x} + \vec{x}^T M \vec{y} + \vec{y}^T M \vec{x} + \vec{y}^T M \vec{y}}$$

$$= \sqrt{\vec{x}^T M \vec{x} + 2\vec{x}^T M \vec{y} + \vec{y}^T M \vec{y}}$$

$$= \sqrt{\|\vec{x}\|_M^2 + 2\vec{x}'^T \vec{y}' + \|\vec{y}\|_M^2}$$

$$\leq \sqrt{\|\vec{x}\|_M^2 + 2\|\vec{x}'\| \|\vec{y}'\| + \|\vec{y}\|_M^2}$$

$$= \sqrt{\|\vec{x}\|_M^2 + 2\|\vec{x}\|_M \|\vec{y}\|_M + \|\vec{y}\|_M^2}$$

$$= \sqrt{(\|\vec{x}\|_M + \|\vec{y}\|_M)^2} = \|\vec{x}\|_M + \|\vec{y}\|_M.$$

$$\text{Let } M = N^T N$$

$$\vec{x}' = N \vec{x}, \vec{y}' = N \vec{y}$$

$$\vec{x}' =$$

$$\|\vec{x}'\| = \|N \vec{x}\|$$

$$= \|\vec{x}\|_M$$

Condition 3 $\forall \vec{x} \in V$ $\sqrt{\vec{x}^T M \vec{x}} \geq 0$

$\sqrt{\vec{x}^T M \vec{x}} = 0$ if $\vec{x} = \vec{0}$ and if $\sqrt{\vec{x}^T M \vec{x}} = 0$, then $\vec{x} = \vec{0}$

Analytic Geometry 2. Distance, angle, orthonormal Basis

Inner product of functions.

Fact: Any inner product induces a norm.

$$\|\vec{x}\| = \sqrt{\vec{x}^T M \vec{x}} \text{ for some sym. positive definite matrix}$$

Def. Distance.

Consider an inner product space $(V, \langle \cdot, \cdot \rangle)$, Then, the distance between any vectors $\vec{x}, \vec{y} \in V$ is

$$d(\vec{x}, \vec{y}) := \|\vec{x} - \vec{y}\| = \sqrt{\langle \vec{x} - \vec{y}, \vec{x} - \vec{y} \rangle}$$

Remark 1: inner product induced a norm, and distance between two vectors is related to the induced norm, i.e. the positive definite matrix M related to the norm.

Remark 2: different norms correspond to different distance

Consider the length of vector $\vec{x} = [1, 1]^T \in \mathbb{R}^2$ using

(1) Dot product. $x_1 y_1 + x_2 y_2$

$$(2) \langle \vec{x}, \vec{y} \rangle := \vec{x}^T \begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix} \vec{y} = x_1 y_1 - \frac{1}{2}(x_1 y_2 + x_2 y_1) + x_2 y_2$$

$$\vec{x} = [x_1, x_2]^T, \vec{y} = [y_1, y_2]^T$$

Remark: length a vector is related to the norm selected, length may be different if norm is different

Def. Metric.

The mapping $d: V \times V \rightarrow \mathbb{R}$ for which $(\vec{x}, \vec{y})$ maps to $d(\vec{x}, \vec{y})$ is called a metric, which satisfies

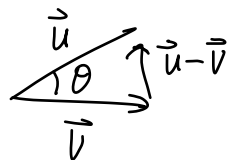
1) positive definite: $d(\vec{x}, \vec{y}) \geq 0$ for any $\vec{x}, \vec{y} \in V$, and $d(\vec{x}, \vec{y}) = 0$ iff $\vec{x} = \vec{y}$

2) symmetric property: $d(\vec{x}, \vec{y}) = d(\vec{y}, \vec{x}) \quad \forall \vec{x}, \vec{y} \in V$

3) triangular inequality: $d(\vec{x}, \vec{z}) \leq d(\vec{x}, \vec{y}) + d(\vec{y}, \vec{z})$

Def Angle

Law of cosine.



$$\| \vec{u} - \vec{v} \|^2 = \| \vec{u} \|^2 + \| \vec{v} \|^2 - 2 \| \vec{u} \| \| \vec{v} \| \cos \theta$$

note that $\langle \vec{u} - \vec{v}, \vec{u} - \vec{v} \rangle = \| \vec{u} \|^2 + \| \vec{v} \|^2 - 2 \langle \vec{u}, \vec{v} \rangle$

$$\Rightarrow \langle \vec{u}, \vec{v} \rangle = \| \vec{u} \| \| \vec{v} \| \cos \theta$$

Assume that $\vec{x} \neq \vec{0}$, $\vec{y} \neq \vec{0}$, then

$$\cos \theta = \frac{\langle \vec{x}, \vec{y} \rangle}{\| \vec{x} \| \| \vec{y} \|}$$

$\theta \in [0, \pi]$ is called the angle between $\vec{x}$, $\vec{y}$.

$\vec{x} \perp \vec{y} \Rightarrow \langle \vec{x}, \vec{y} \rangle = 0 \Rightarrow \vec{x}$ and $\vec{y}$ are orthogonal.

Furthermore, if $\vec{x}$ and $\vec{y}$ are orthogonal and $\| \vec{x} \| = \| \vec{y} \| = 1$ then $\vec{x}$ and $\vec{y}$ are both orthonormal.

Def. Orthogonal Matrix.

A square matrix $A \in \mathbb{R}^{n \times n}$ is an orthogonal matrix iff its columns are orthonormal.

$$A^T A = A A^T = I \text{ implies } A^{-1} = A^T$$

$\Rightarrow$ inverse is obtained by simply transposing the matrix.

$$A^T A = \begin{bmatrix} \vec{a}_1^T \\ \vec{a}_2^T \\ \vdots \\ \vec{a}_n^T \end{bmatrix} [\vec{a}_1 \cdots \vec{a}_n] = \begin{bmatrix} \vec{a}_1^T \vec{a}_1 & \vec{a}_1^T \vec{a}_2 & \cdots & \vec{a}_1^T \vec{a}_n \\ \vec{a}_2^T \vec{a}_1 & \vec{a}_2^T \vec{a}_2 & \cdots & \vec{a}_2^T \vec{a}_n \\ \vdots & \vdots & \ddots & \vdots \\ \vec{a}_n^T \vec{a}_1 & \vec{a}_n^T \vec{a}_2 & \cdots & \vec{a}_n^T \vec{a}_n \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

Remark: Transformations by orthogonal matrices do NOT

change the length of a vector

$$\| A \vec{x} \|^2 = (A \vec{x})^T A \vec{x} = \vec{x}^T A^T A \vec{x} = \vec{x}^T I \vec{x} = \vec{x}^T \vec{x} = \| \vec{x} \|^2$$

self learning: orthonormal basis (sec 3-5)
orthonormal complement (sec 3-6)

★ Inner product of Functions.

Given two functions, $u, v: \mathbb{R} \mapsto \mathbb{R}$, the inner product of u and v is defined as $\langle u, v \rangle := \int_a^b u(x)v(x)dx$, with $a, b < \infty$

A generalization of inner product to infinite number of entries

Remark: norm and orthogonality of a function (among functions) can also be define in a similar way (sum $\rightarrow$ integral)

Related concept: convolution of two function.

$$u * v(t) = \int_0^t f(\tau)g(t-\tau)d\tau$$

$\rightarrow$ convolutional neural networks $\rightarrow$ image recognitions

Summary:

